

Project Design Phase

Proposed Solution

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict Record Access Based On Field Value
Maximum Marks	2 Marks

Proposed Solution Matrix

Project team framework details outlining parameters for mapping custom field values against security execution contexts:

S.No.	Parameter	Description
1.	Idea / Solution description	This solution proposes a clean context-aware database wrapper and microservice middleware engine governed via a dynamic Script-Controlled Access Control List (ACL). Whenever an application requests records, a database interceptor layer catches the query execution tree before dispatching raw payloads. It safely injects field validation scripts executed inside a sandbox engine. The platform parses the runtime row metrics (e.g., verifying if Document.FieldStatus == 'Restricted') alongside active session JSON tokens, seamlessly stripping out unauthorised columns or entire indices automatically without altering the primary business logic flow.
2.	Novelty / Uniqueness	The key technical differentiator is decoupling data-level visibility scripting entirely from native application source code and direct engine query filters. Administrators can write, manage, run sandbox unit tests, and deploy updated javascript expressions or custom script

		logic blocks in real-time straight through a centralized security operational panel. The middleware dynamically re-evaluates security configurations hot-swapped over instances with absolutely zero compilation overheads, service downtime, or repository pull-request cycles.
3.	Social Impact / Customer Satisfaction	Provides robust data privacy compliance structures that inherently protect sensitive personal data records from internal insider misuse and misconfigured cloud endpoints. For corporate customers, tech platforms, and regulatory compliance teams, it translates to clean automated compliance verification trails, zero administrative fatigue handling policy overhauls, and reduced risks of severe enterprise data exposure leaks.
4.	Business Model (Revenue Model)	Dual Enterprise Monetization Vector Layout: • Core Developer Engine Open-Source: Distributed under an MIT/Apache core module license to quickly encourage community adoption and native framework ecosystem expansions. • Commercial Managed SaaS / Self-Hosted Enterprise Subscription: Includes managed cloud integration utilities, real-time logging telemetry dashboards, visual sandboxed script editors, high-availability horizontal failover controllers, and priority operational SLA assistance contracts.
5.	Scalability of the Solution	The script evaluation core is intentionally built as a stateless middleware element capable of seamless horizontal scaling

		behind cluster load balancers across multi-region Kubernetes groups. To eliminate latency bottlenecks during complex database record queries, the system introduces a two-tier high-performance architecture: caching compiled logic structures inside fast Redis lookup stores and optimizing structural field checks with specialized pre-filtering query compilation scripts.
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